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International Mathematics Assessments for Schools

2021 ~ 2022 Junior Division 2nd Round Paper

Time: 120 minutes

Printed Name _____ *Code* _____ *Score* _____

Instructions:

- Do not open the contest booklet until you are told to do so.
- Be sure that your name and code are written on the space provided above.
- Round 2 of IMAS is composed of three parts; the total score is 100 marks.
- Questions 1 to 5 are given as a multiple-choice test. Each question has five possible options marked as A, B, C, D and E. Only one of these options is correct. After making your choice, fill in the appropriate letter in the space provided. Each correct answer is worth 4 marks. There is no penalty for an incorrect answer.
- Questions 6 to 13 are a short answer test. Only Arabic numerals are accepted; using other written text will not be honored or credited. Some questions have more than one answer, as such all answers are required to be written down in the space provided to obtain full marks. Each correct answer is worth 5 marks. There is no penalty for incorrect answers.
- Questions 14 and 15 require a detailed solution or process in which 20 marks are to be awarded to a completely written solution. Partial marks may be given to an incomplete presentation. There is no penalty for an incorrect answer.
- Use of electronic computing devices is not allowed.
- Only pencil, blue or black ball-pens may be used to write your solution or answer.
- Diagrams are not drawn to scale. They are intended as aids only.
- After the contest the invigilator will collect the contest paper.

The following area is to be filled in by the judges;
the contestants are not supposed to mark anything here.

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Total Score	Signature
Score																	
Score																	

2021 ~ 2022 Junior Division 2nd Round Paper

Questions 1 to 5, 4 marks each

1. Let x and y be non-zero real numbers such that

$$\begin{cases} x = y + \dfrac{4}{x} \\ y = x + \dfrac{5}{y} \end{cases}.$$

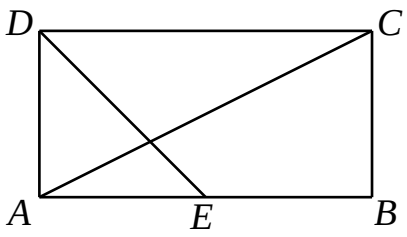
Find the sum of all possible values of $|x - y|$.

（A）4　　（B）5　　（C）9　　（D）3　　（E）0

Answer：_____

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2. In the diagram below, $ABCD$ is a rectangle, where AB = 8 cm and BC = 4 cm. Let the angle bisector from D intersect AB at E.



Find the distance, in cm, from E to AC.

（A）$\dfrac{3\sqrt{2}}{2}$　　（B）$\dfrac{4\sqrt{2}}{5}$　　（C）$\dfrac{4\sqrt{5}}{5}$　　（D）$\dfrac{2\sqrt{5}}{5}$　　（E）$\dfrac{4\sqrt{2}}{5}$

Answer：_____

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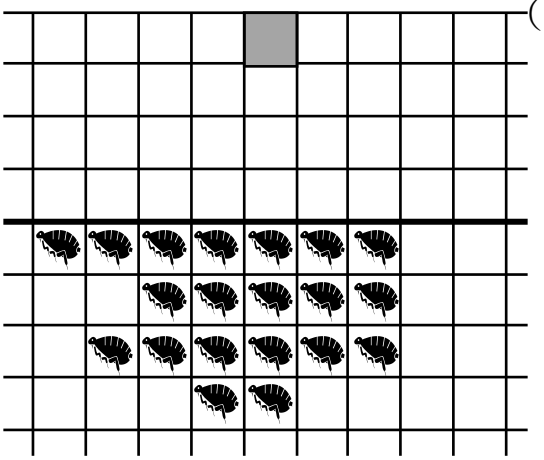
3. Using the set of integers from 1 to 2022 inclusive, at most how many integers can we choose such that the positive difference between any two chosen numbers is not a prime number?

（A）253　　（B）254　　（C）504　　（D）505　　（E）506

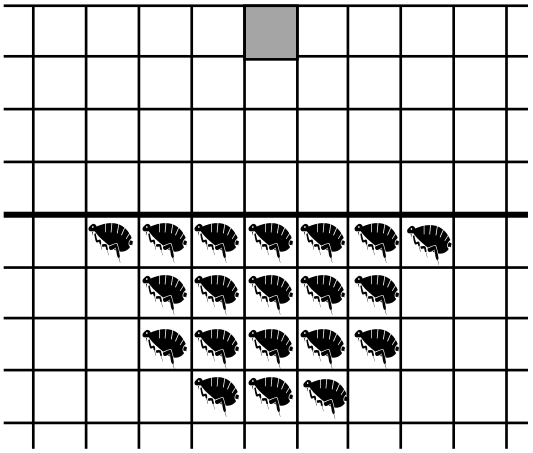
Answer：_____

4. The world of the beetles consists of one entire plane divided into unit squares. Initially, all the beetles are located at squares south of an inner wall and each square is occupied by at most one beetle. In each move, a beetle can jump over another beetle in an adjacent square and land on the square immediately beyond. The beetle that was jumped over is then removed. However, the move is not permitted if the square is already occupied. The jump may be northward, eastward or westward. How can one arrange 20 beetles such that they can reach the shaded square in the diagram below?

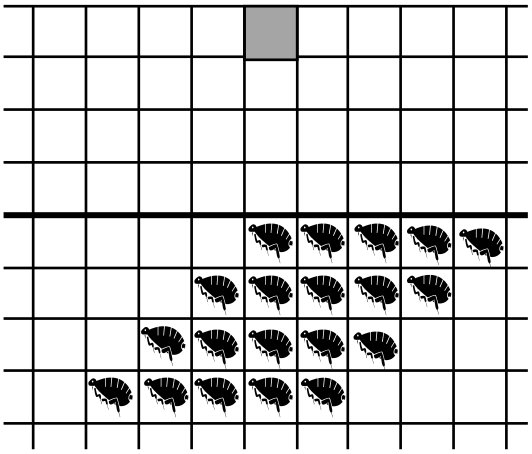
(A)



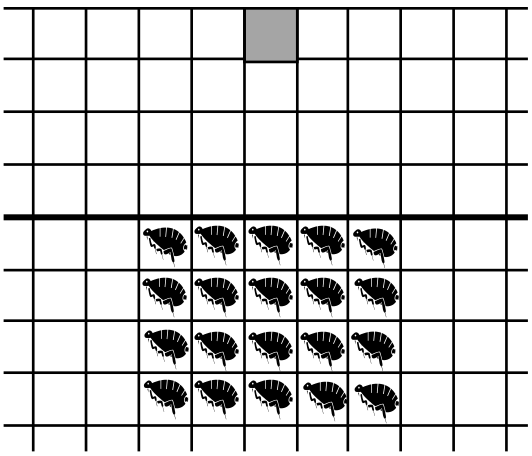
(B)



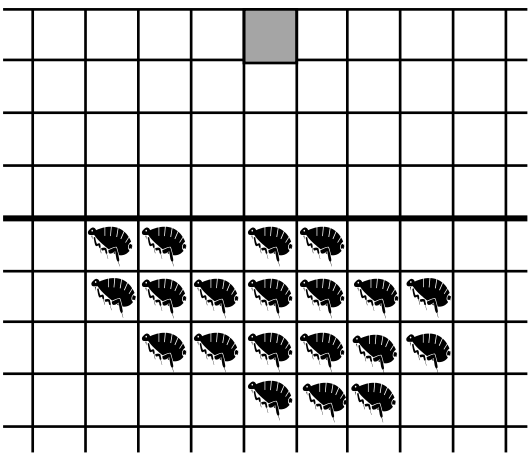
(C)



(D)

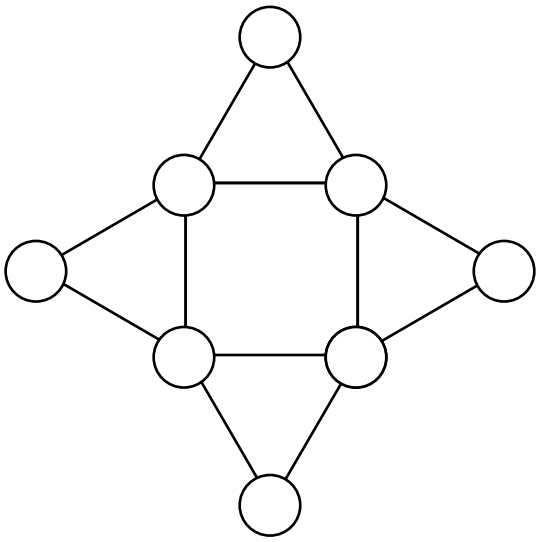


(E)



Answer : _____

5. Numbers 1, 2, 3 and 4 are to be placed into the circles in the diagram shown, where each circle must be filled with exactly one number. If two circles connected by a line should not contain the same number, find the total number of ways in filling the diagram.



（A）2016　（B）1344　（C）672　（D）576　（E）384

Answer：_____

Questions 6 to 13, 5 marks each

6. We define: $\boxed{2}=1\times2\times3$, $\boxed{3}=2\times3\times4$, $\boxed{4}=3\times4\times5$, $\boxed{5}=4\times5\times6$, ….

If $\frac{1}{\boxed{6}}-\frac{1}{\boxed{7}}=\frac{1}{\boxed{7}}\times A$, what is the value of A?

Answer：_____

7. What is the value of the irreducible fraction $\frac{n}{m}$, where $0.17<\frac{n}{m}<0.18$ and m is as small as possible?

Answer：_____

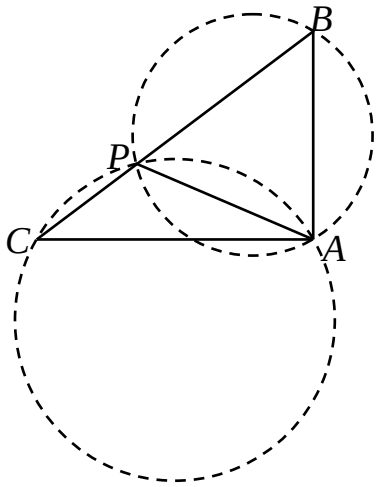
8. Define $f(x)=\frac{1}{\sqrt[3]{x^2+2x+1}+\sqrt[3]{x^2-1}+\sqrt[3]{x^2-2x+1}}$, find the value of $f(1)+f(3)+...+f(2k-1)+f(999)$.

Answer：_____

9. What is the last non-zero digit of 2022! ?

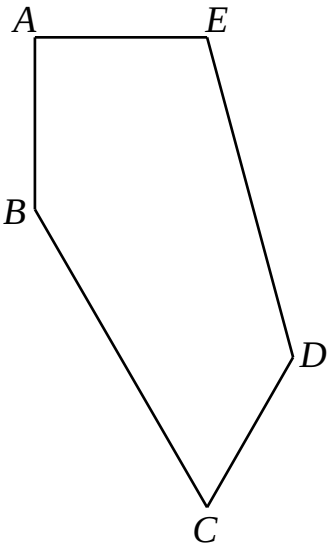
Answer：_____

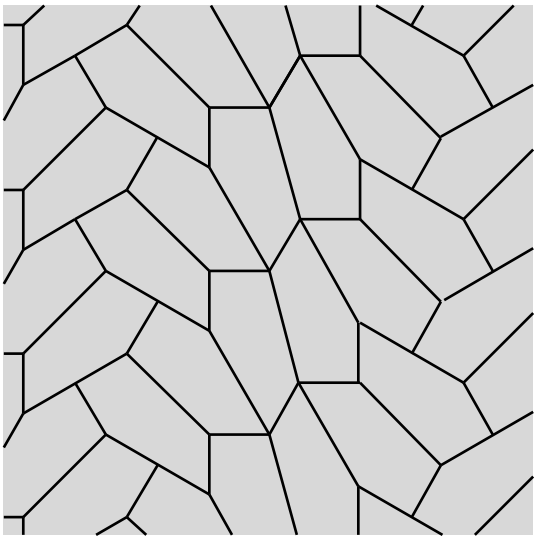
10. Let ABC be a triangle, where $AB = 12$ cm, $AC = 16$ cm and $BC = 20$ cm. If point P lies on segment BC, what is the minimum sum of the circumradii, in cm, of triangles PAB and PAC?



Answer：_____ cm

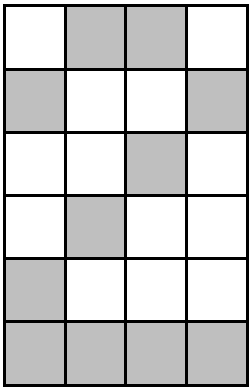
11. As shown in the diagram below, we use some number of pentagons, that is identical to $ABCDE$ to fill the plane below. Suppose AB is 1 cm, what is the side length, in cm, of DE?





Answer：_____ cm

12. Some unit squares of a 6×4 board are coloured gray, as shown in the diagram below. Find the number of ways in choosing four different gray unit squares such that any two unit squares that are chosen are located on different rows and different columns.



Answer : _____ ways

13. In a purchase, the quantity of item *X* must be not more than twice of quantity of item *Y*, but not less than one third of quantity of item *Y*. Moreover, the total quantity of the two products must not be more than 60. Find the total number of combinations of non-zero quantities satisfying these requirements.

Answer : _____

Questions 14 to 15, 20 marks each
(Detailed solutions are needed for these two problems)

14. Seven integers a_0, a_1, …, a_6, where a_i is the number of *i*'s among a_0, a_1, …, a_6, for each $0 \le i \le 6$. Find the value of these numbers.

Answer : _____

15. It is known that positive numbers x, y and z satisfy

$$\begin{cases} x^2 + xy + y^2 = 5^2 \\ y^2 + yz + z^2 = 12^2 \\ z^2 + xz + x^2 = 13^2 \end{cases}$$

What is the value of $(x + y + z)^2$?

Answer：_____